



2.3

Exponential Functions

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 2

| ECHS Mathematics

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Student Information

Name: _____ Class: _____ Date: _____

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: For $b > 1$, b^x is increasing; for $0 < b < 1$, it is decreasing.

2. Write one precise sentence explaining this principle: The parent exponential has domain all real numbers, range positive numbers, and horizontal asymptote $y = 0$.

3. Write one precise sentence explaining this principle: A vertical shift moves the horizontal asymptote.

4. Write one precise sentence explaining this principle: Nonzero exponential functions do not have x-intercepts unless shifted.

Core

5. Analyze $f(x) = 3(1/2)^{x-2} - 4$.
6. Find the growth factor if an exponential doubles every 5 units.
7. Can $2^x + 1$ equal 0?
8. State what the formula $f(x) = ab^{x-h} + k$ tells you and name any required restriction.
9. State what the formula horizontal asymptote $y = k$ tells you and name any required restriction.
10. State what the formula $\frac{f(x+1)-k}{f(x)-k} = b$ tells you and name any required restriction.

Medium

11. Analyze domain, range, asymptote, monotonicity, and concavity of $3 \cdot 2^x$.
12. Analyze domain, range, asymptote, monotonicity, and concavity of $5(1/3)^x$.
13. Analyze domain, range, asymptote, monotonicity, and concavity of $-2^x + 4$.
14. Analyze domain, range, asymptote, monotonicity, and concavity of $2 \cdot 3^{x-1} - 5$.

15. Analyze domain, range, asymptote, monotonicity, and concavity of $-4(1/2)^{x+2} + 1$.

Hard

16. Find a formula $f(x) = ab^x$ through (0,6) and (2,24).

17. Find x-intercept of $3 \cdot 2^x - 12$.

18. For $f(x) = ab^x + k$, show $(f(x+1) - k)/(f(x) - k) = b$.

19. Determine whether $f(x) = 2^{-x}$ is increasing or decreasing and rewrite it.

Difficult

20. Find parameters in $f(x) = a2^x + k$ given horizontal asymptote $y=-3$ and points (0,1),(1,5).

21. Find all b for which b^x is its own reflection across the y -axis.

22. Compare end behavior of 2^x and x^{10} .

Challenge / HOT

23. Prove a nonconstant exponential ab^x has no repeated output when b is positive and not 1.

24. Construct two different exponential forms representing the same function.

AP-Style Multiple Choice

25. Base b for exponential must satisfy

- (A) $b < 0$
- (B) $b > 0, b \neq 1$
- (C) $b = 1$
- (D) b integer only

26. Parent exponential range is

- (A) $\mathbb{R}$
- (B) $(0, \infty)$
- (C) $[0, \infty)$
- (D) $(-\infty, 0)$

27. Horizontal asymptote of $2^x - 7$ is

- (A) $x=-7$
- (B) $y=-7$
- (C) $y=0$
- (D) $x=0$

28. $(1/3)^x$ is

- (A) increasing
- (B) decreasing
- (C) constant
- (D) not a function

29. Initial value of $5 \cdot 1.2^x$ is

- (A) 1.2
- (B) 5
- (C) 6
- (D) 0

30. Exponential parent is always

- (A) concave up
- (B) concave down
- (C) linear
- (D) bounded above

AP-Style Free Response

31. Let $f(x) = ab^{x-h} + k$ have horizontal asymptote $y=2$, pass through (1,5) and (3,14), and $b>1$. Choose $h=1$. Find a, b, k and analyze.

32. A transformed exponential has range $(-\infty, 6)$, is increasing, and passes through (0,2),(1,4). Construct one formula.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.